

Exact optimal strategies for the WordHunt oracle-query game

A research logbook (Volume 1: getting started)

2026-09-17 – 2026-09-18

ARCHIVED 2026-09-18. This is the historical record of how this project got started: the first attempt at a generative model (called M_1 here) turned out not to be a valid model of the real game at all, and the oracle behaviour first coded was simply wrong. Both errors, and several downstream consequences of each, are found and corrected *in place, over the course of this document* – meaning large parts of it, including some of its own “corrected” sections, state things that are later shown to be wrong again. That is deliberate: this volume is kept as a faithful record of the process, not as a reference.

Do not use this document as a source of current facts about the game, the model, or any numeric result. For a correct, self-contained statement of the game, the model actually used ($M_2 - M_1$ does not appear in it at all, having been found not to be a valid model), the objective, and every result that survives review, see `research/logbook.tex` (Volume 2), which supersedes this document entirely.

1 Motivation

`wordhunt.py` / `wordhunt2.py` (in the parent directory) are recreational assistance tools for a daily puzzle: 25 letters (one letter of the alphabet entirely absent) fill a 5×5 grid; a single 5-letter dictionary word is spelled along one of 12 lines (5 rows, 5 columns, 2 diagonals); the player learns, one query at a time, where a chosen letter actually sits in the grid (the grid is a full bijection over every non-excluded letter, so this is disclosed regardless of whether that letter turns out to be part of the answer word).

Every statistic those tools compute — raw letter frequency, Zipf-weighted frequency, positional entropy, a proposed Bayesian “leading candidate” posterior — turned out, on inspection, to be a *one-step heuristic* resting on an unstated and unverified assumption about how the grid was generated. Repeated design discussion (2026-09-17) concluded that heuristics of this kind cannot be validated or compared meaningfully until at least one fully rigorous baseline exists: a strategy that is *provably optimal* for an *explicitly stated* generative model M and an *explicitly stated* objective L .

This logbook is that project. The plan (agreed 2026-09-17) is:

1. Shrink the game to a size where every board, and every possible play-through, can be exhaustively enumerated by brute force — so every claim can be machine-checked, not just argued.
2. Get a genuinely optimal solver working and validated there.

3. Scale up board size step by step, tracking exactly where and why brute-force techniques stop being tractable, and look for workarounds at each wall before conceding defeat.
4. Keep this document updated as a persistent record, so a negative result (“exact solving becomes impractical beyond size k , for reason X”) is as valid and as carefully written up as a positive one.

All code lives in `research/` on the `optimal-solver-research` git branch, kept separate from the recreational tools so nothing here can break them. All numeric results quoted below were produced by the code in this directory and are reproduced by `research/tests/`.

Notice to the reader (added after Section 11 was written)

Sections 2–9 below were written *before* two real bugs were found later the same day: a wrong oracle model, and (downstream of fixing that) a retracted sufficiency lemma. The model/algorithm descriptions in Sections 2–4 have been corrected in place to describe the model actually intended (and now actually implemented) rather than left wrong for narrative purposes – a documentation bug is not historical data worth preserving the way a numeric result is. The sufficiency lemma (Section 5) and every numeric-results section built on it (Sections 6–9) are marked **SUPERSEDED** at the point where they appear, with a pointer to Section 11, rather than silently rewritten or deleted, per this logbook’s stated practice. Read Section 11 before trusting any number before it.

2 Model M_1 : the generative process

Parametrised by board size n (the real game is $n = 5$):

- Alphabet size = $n^2 + 1$ (exactly one letter of the alphabet is absent from every grid, mirroring the real game).
- Grid is $n \times n$; **lines** (candidate word positions) are the n rows, n columns, and 2 diagonals: $2n + 2$ in total.
- A fixed vocabulary W of length- n strings, each with n distinct letters (not required to be English — just a fixed toy dictionary for small n ; the real `dict_game.csv` is used once $n = 5$).

M_1 generates a grid as:

1. word $w \sim \text{Uniform}(W)$
2. line $p \sim \text{Uniform}(\text{the } 2n + 2 \text{ lines})$
3. excluded letter $e \sim \text{Uniform}(\text{alphabet} \setminus w)$
4. the remaining $n^2 - n$ letters are placed in the remaining $n^2 - n$ cells via a uniformly random permutation

with *no rejection step*. M_1 is deliberately the simplest well-posed model, and is known to be wrong in at least one respect: the 2026-09-17 discussion raised the possibility that a real puzzle setter rejects (and retries) any grid whose cross letters accidentally spell a second valid word along some other line. That complication is deferred to a future M_2 ; every result below is explicitly a result *about* M_1 , not a claim about the real puzzle.

The excluded letter e is told to the player before play starts (as in the real game), so it is fixed per game instance rather than left as a live hypothesis.

2.1 Oracle

Querying an untested letter $L \neq e$ deterministically returns *the one cell where L actually sits in the grid* – whether or not L turns out to be part of the answer word. If L happens to be part of the answer word, that cell is necessarily $p[w.\text{index}(L)]$; if not, it is whichever of the $n^2 - n$ dross cells the generative process (step 4 above) happened to place L at. There is no “not in the word, no location” outcome: the grid is a full bijection over every non-excluded letter, so every query resolves to a real cell. This matches `wordhunt.py`’s actual bookkeeping – every entry in its `revealed` dict is handled identically by `word_fits_position` regardless of whether that letter later turns out to be part of the answer.

(An earlier draft of this section, and the first implementation of `query_outcome` in `solver.py`, stated and coded the opposite – that a dross letter’s query returns no location at all. That was a real bug, found and fixed the same day; see Section 11 for how it was caught and what it broke downstream.)

3 Objective L_1

Minimise $\mathbb{E}[\text{number of queries until the grid is uniquely determined}]$ (equivalently, until (w, p) is uniquely determined, in every state reachable without a collision — see Section 11 on why grid uniqueness, not (w, p) uniqueness, is the well-posed target once M_1 has no veto step, and on the still-open question of whether this is stricter than the real game’s actual win condition).

4 Exact DP

Let H be the current set of *candidate worlds* (full grids) consistent with everything observed so far. Under M_1 , every candidate world is equally likely, so:

$$V(H) = 0 \quad \text{if every remaining candidate has the same grid} \quad (1)$$

$$V(H) = 1 + \min_L \sum_{\text{cell } c} \frac{|H_c|}{|H|} V(H_c) \quad \text{otherwise} \quad (2)$$

where $H_c \subseteq H$ is the subset of candidates in which L actually sits at cell c (Section 2.1 on why this partitions by concrete cell, with no separate merged “ L absent” bucket: the oracle always discloses L ’s true cell). Termination is on grid equality, not on agreement of some tagged (w, p) pair — see Section 11 on why the latter can be an unanswerable question when M_1 has no veto step.

The $\arg \min_L$ at the root is the provably optimal next query — not a proxy for it. This is implemented as `ExactSolver`, operating on fully-enumerated worlds, in `solver.py`. (A second solver, `PairSolver`, was written to operate directly on (w, p) pairs without full worlds, on the strength of a sufficiency argument in Section 5 below – that argument is **retracted**; see Section 11. `ExactSolver` on full worlds is the only currently-trusted solver.)

Also implemented: `greedy_choice`, the one-step maximum-information-gain heuristic (i.e. choosing L to maximise the Shannon entropy of the outcome partition on its own, ignoring everything beyond one query) — the same style of proxy used throughout `wordhunt2.py`. Comparing it against `ExactSolver` is the central point of this exercise.

5 The (word, line) sufficiency lemma

RETRACTED (see Section 11). The lemma below is FALSE under the corrected oracle of Section 2.1: a dross letter’s true cell depends on the specific dross permutation, which is not determined by (w, p) alone, so (w, p) pairs are not in fact a sufficient statistic. It held only under the earlier, buggy oracle (where a dross query carried no location information at all, so there was nothing about the permutation left to depend on). **PairSolver**, and every result below and in Sections 8–9 that relies on it, are therefore also unsound as stated. Kept below, unmodified, as the argument that was believed correct at the time and the historical verification that was run against it – not as a currently-trusted claim.

Full-world enumeration (build every literal grid M_1 can produce) is combinatorially impossible beyond very small n : the number of ways to permute the $n^2 - n$ dross letters into the remaining cells is $(n^2 - n - 1)!$.

n	alphabet size	lines	dross permutations $(n^2 - n - 1)!$
2	5	6	$2! = 2$
3	10	8	$6! = 720$
4	17	10	$12! = 479,001,600$
5	26	12	$20! \approx 2.43 \times 10^{18}$

Lemma. Under M_1 , with the excluded letter e fixed, every (w, p) hypothesis has *exactly* the same number of full-world realisations (the dross-permutation count depends only on n , not on which word or line), so this factor is a constant that cancels out of every relative-probability calculation. The exact DP can therefore be run directly on (w, p) pairs, weighting each equally, without ever constructing a full grid.

Verification (as it stood before the fix – now invalid, see the retraction notice above). Rather than trust the algebra alone, `tests/test_pair_solver.py` built both **ExactSolver** (on fully-enumerated worlds) and **PairSolver** (on pairs only) for every excluded-letter scenario at $n = 2$ and $n = 3$ and asserted the two reported bit-identical $V(H_0)$; under the buggy oracle then in use, both did pass. Once the oracle was corrected, the same tests were re-run and now fail (kept in the suite as `xfail/skip`, deliberately, per Section 11) – the “verification” this paragraph describes verified the wrong model self-consistently, not the real one.

Had it remained valid, this would have removed the dross-permutation factorial as an obstacle entirely, independent of the DP’s own combinatorial cost (the separate wall in Section 9, also now superseded). That structural point — factorial-avoidance and DP-cost are two distinct obstacles — still stands even though the specific lemma that was going to deliver the first one does not.

6 Results at $n = 2$

SUPERSEDED (see Section 11). Computed under the buggy oracle of Section 2.1’s earlier draft. The corrected re-solve is in Section 11 (§11, “Re-solved: $n = 2$ ”); the values below (2.2778) are not the corrected ones (2.944) and should not be quoted. Kept for the historical comparison in Finding 1 below, which is itself superseded by Section 11’s discussion of why it changed.

Toy vocabulary (alphabet $\{a, b, c, d, e\}$): $W = \{ab, bc, cd, de, ea\}$ (deliberately cyclic/symmetric).

excluded	candidate words	optimal $\mathbb{E}[\text{queries}]$	greedy $\mathbb{E}[\text{queries}]$
a	3	2.2778	2.2778
b	3	2.2778	2.2778
c	3	2.2778	2.2778
d	3	2.2778	2.2778
e	3	2.2778	2.2778

Finding 1. At $n = 2$, greedy exactly matches the true optimum in every scenario tested. This looks like a validation of the greedy heuristic, but Section 7 shows it is an artefact of $n = 2$ being too small/symmetric a case for greedy’s known failure mode to appear — not evidence that greedy is generally optimal.

Every $\mathbb{E}[\text{queries}]$ figure above was independently cross-checked by `simulate_policy_exact_expectation`, an independently-coded exact recursion that replays the solver’s chosen policy against every one of the 36 enumerated worlds per scenario and recomputes the same expectation from scratch (`test_exact_solver_value_matches_independent_simulation`).

7 Results at $n = 3$: greedy is provably suboptimal

SUPERSEDED (see Section 12.5). Computed under the buggy oracle. Re-verified under the corrected oracle 2026-09-18: contrary to the 2026-09-17 entry’s claim that `ExactSolver` “did not finish within 20 seconds” and that this was therefore unknown, it does finish – in 26–42s, at 760k–1.17M memoised states, once actually given enough time rather than a 20s budget chosen without checking. That 2026-09-17 claim was itself an error (impatience, not intractability) and is corrected in Section 12.5, which also confirms greedy-suboptimality holds in *all ten* scenarios under the corrected oracle and M_2 , not just 6 of 10 as below. The qualitative point (greedy $\neq$ optimal in general) survives either way.

Toy vocabulary (alphabet $\{a..j\}$): $W = \{\text{abc, cde, efg, ghi, ijb, bad, dec, fah}\}$.

excluded	candidate words	worlds	optimal	greedy
a	5	28800	2.3000	2.3000
b	5	28800	2.3250	2.3250
c	5	28800	2.5000	2.5500
d	5	28800	2.5250	2.5500
e	5	28800	2.3250	2.3250
f	6	34560	2.5000	2.5417
g	6	34560	2.5000	2.5417
h	6	34560	2.5000	2.5000
i	6	34560	2.5208	2.5417
j	7	40320	2.6964	2.7321

Finding 2 (the headline result so far). In 6 of the 10 scenarios, one-step maximum-entropy greedy selection is *strictly* worse than the true optimum — not by a rounding error, but by a consistently reproducible margin (e.g. excluded=‘c’: 2.55 vs. 2.50). This is an exact, exhaustively-verified counterexample to trusting greedy one-step information gain as “the” optimal-play criterion, of exactly the kind known from optimal Mastermind/Wordle solvers: two letters can tie (or nearly tie) on immediate information gain while differing in how good a position they leave for the *next*

query. Every number above is exact (not simulated) and was solved in under 1.5s per scenario (`research/tests`, and the ad-hoc script in the git history of this branch).

This directly undercuts every “pick the highest-entropy/highest-Bayes-mass letter” heuristic used by `wordhunt2.py` to date: even granting their weighting assumptions, one-step greedy selection on top of them is demonstrably not the same thing as optimal play.

8 The $n = 4$ wall (diagnosed, and defused)

SUPERSEDED (see Section 11). This section’s conclusion (“thanks to the sufficiency lemma, this does not matter”) relies entirely on the retracted lemma of Section 5. The cross-permutation factorial obstacle described below is real and correctly diagnosed; the claimed resolution (via `PairSolver`) is not currently valid. $n = 4$ should be considered unaddressed again until a correct pair-based (or otherwise sub-factorial) solver exists.

Full-world enumeration is impossible at $n = 4$ (Section 5’s table: $12! \approx 4.8 \times 10^8$ cross permutations per (word, line, excluded) choice). Thanks to the sufficiency lemma, this does not matter: `PairSolver` needs no full worlds, and its state space at $n = 4$ is just (number of candidate words) $\times$ (10 lines) – no factorial term at all. The real question is whether *that* state space is tractable, which turns out to depend overwhelmingly on n (alphabet/branching factor) and vocabulary size rather than on the factorial term that looked scary at first glance.

9 The real wall: $n = 5$ with realistic vocabulary size

SUPERSEDED (see Section 11). Every number below was computed with `PairSolver`, which is now known to silently implement the wrong (retracted) oracle model throughout this whole section, not just at the edges – so these are not a measurement of M_1 ’s real difficulty, they are a measurement of a different, incorrect game’s difficulty. The qualitative shape of the finding (branching factor in n drives a combinatorial wall well short of realistic dictionary size) is plausible and worth re-measuring once a correct solver exists, but the specific numbers below (e.g. “40 words timed out”) must not be quoted as a property of the real game or of M_1 until redone.

Using `PairSolver` (validated, not full-world enumeration) directly on random samples of real English words from `dict_game.csv` (excluded letter fixed at ‘h’, matching a real play session from 2026-09-17), starting from the fully-uninformed state:

sample size (words)	(w, p) pairs	optimal $\mathbb{E}[\text{queries}]$	wall-clock	DP states memoised
10	120	2.583	0.08 s	896
15	180	2.894	0.54 s	3,441
20	240	3.108	2.57 s	11,525
25	300	3.337	5.66 s	23,925
30	360	3.381	10.04 s	39,725
40	480	—	>25 s, incomplete	>58,237

Memoised-state growth is well above linear in the number of pairs (roughly consistent with $O(|H_0|^{3.5})$ across this range, though the sample is too small to fit a reliable exponent): doubling the pair count from 180 to 360 grows the state count by $\sim 11.5\times$, not $2\times$.

Finding 3. The real `dict_game.csv` has 10,176 candidate words, i.e. roughly $300\times$ the 30-word sample that already took 10 seconds and 40,000 memoised states. Extrapolating even the optimistic

end of the observed growth curve, exact full-lookahead solving at real $n = 5$ scale, from the fully-uninformed opening state, is not a performance-tuning problem — it is many orders of magnitude beyond reach on this hardware, and almost certainly on any hardware, without a fundamentally different algorithmic approach (see Section 10).

Importantly, this wall is a property of n (i.e. the branching factor: up to 21 queryable letters at $n = 5$ vs. up to 7 at $n = 3$) and of how *early* in the game the search starts (the fully-uninformed opening state has the largest candidate set the game ever sees), much more than it is a property of vocabulary size on its own: Section 7’s $n = 3$ instances, with comparable vocabulary sizes (5–8 words) but a 6–7-letter query alphabet instead of a 21-letter one, solved in well under a second.

10 Next steps

Not yet attempted, in rough order of expected value:

1. **Branch-and-bound with a provable lower bound.** A node can be pruned the moment its best-case remaining cost (e.g. log of the branching factor) already exceeds the best solution found so far. This preserves exactness (no silent fallback to a heuristic) while potentially cutting large amounts of dead search.
2. **Solve from mid-game states, not the opening.** A real player only needs an exact answer from *wherever they currently are* – typically well past the opening, with the candidate set already substantially pruned by revealed cells. Section 9’s wall is specifically about the worst case (opening, full 10k-word dictionary); it may still be exactly solvable a few queries in, once $|H|$ has naturally shrunk. Worth measuring directly rather than assuming.
3. **Depth-limited exact search with a reported bound.** Search exactly to some fixed depth d , and for states beyond that, report a provable interval (not a guess) rather than a heuristic point estimate – matches the “never bluff a heuristic under the optimal-solver label” principle from the design discussion.
4. **Model M_2 :** reintroduce the “dross letters must not accidentally spell a second word” rejection step raised on 2026-09-17, and check whether M_1 ’s uniform-pair-weighting result changes qualitatively once that’s modelled, at whatever board size remains tractable.
5. **A different objective L_2 :** $\mathbb{E}[(\text{queries})^2]$ requires tracking two moments through the Bellman recursion instead of one; algebraically routine but not yet implemented.

SUPERSEDED (see Section 11). The paragraph below asserted “verified at toy scale ($n \leq 3$)” – that is no longer established: $n = 3$ under the corrected oracle has not been observed to terminate. Only $n = 2$ is currently verified. The shape of the argument (exact play is achievable at small scale, uncertain at $n = 5$, greedy is not a substitute) may still be right, but should be read as a hypothesis to re-check, not a settled conclusion.

It is entirely possible that the honest endpoint of this project is a short paper whose conclusion is: exact optimal play is achievable and verified at toy scale ($n \leq 3$), is real and worth verifying at $n = 5$ from *sufficiently-advanced* mid-game states, but is fundamentally out of reach for the opening move against a realistic dictionary – and greedy one-step heuristics, however dressed up, are not a substitute for it (only a proxy of unknown and state-dependent quality, per Section 7). That would be a complete and legitimate result, not a failure to reach one.

11 Corrections (2026-09-17, same day)

Per this logbook’s stated practice, errors found in the results above are recorded here rather than silently rewritten. **Every numeric result in Sections 6–9 was computed under a wrong oracle model** and should be treated as superseded pending re-derivation.

11.1 Bug 1: the oracle model was wrong

The original `query_outcome` returned *no location at all* when a queried letter was not part of the answer word. This is not what the real game does, and not what `wordhunt.py` implements: the grid is a full bijection over every non-excluded letter, so querying any letter always resolves to its one true cell, whether or not that letter turns out to be part of the answer. A “not in the word” outcome never occurs as a distinct case in `wordhunt.py`’s bookkeeping – every entry in `revealed` is handled identically by `word_fits_position` regardless of whether it later turns out to be dross. `query_outcome` has been corrected to always return the letter’s true cell.

This is the same category of mistake as the earliest abandoned “F(L|G)” attempt from the original design discussion (modelling where a dross letter “would be”), recurring in a new piece of code – worth remembering next time an oracle-behaviour assumption is written down without a test against brute-force enumeration first.

11.2 Bug 2: the stopping condition was ill-posed without a veto step

Fixing bug 1 immediately exposed an infinite loop in the DP. Diagnosis: at $n = 2$, `excluded='e'`, two *different* generative paths – $w = \text{“ab”}$ along the top row, and $w = \text{“cd”}$ along the bottom row – can produce the *identical* grid:

```
World A: word='ab', line=top row grid={(0,0):a,(0,1):b,(1,0):c,(1,1):d}
World B: word='cd', line=bottom row grid={(0,0):a,(0,1):b,(1,0):c,(1,1):d}
SAME GRID? True
```

No sequence of oracle queries can ever tell these apart, because the oracle’s answers are a pure function of the grid, and the grid is identical. The DP’s termination check (“is the tagged (word,line) unique?”) was asking an unanswerable question and looped forever trying to resolve it. This is exactly the concern raised when M_1 ’s lack of a veto step was agreed to: “*if you choose to do that you must then be clear about the stopping condition.*” It surfaced as a literal bug rather than staying a hypothetical.

Fix: termination is now defined on *grid* uniqueness, not tagged-pair uniqueness – i.e. M_1 ’s well-defined notion of “solved” is “the grid is fully reconstructed,” which coincides with “the designated answer is known” except exactly in a collision, where the latter is genuinely unknowable and the former is the most that can honestly be claimed.

Measured collision rate (fraction of raw, unvetoes M_1 worlds whose grid happens to spell more than one dictionary word somewhere):

n	vocabulary size	collision rate
2	5 words	69% (all excluded-letter scenarios)
3	5–7 words	4.3%–8.2%, depending on excluded letter

The $n = 2$ figure is inflated by an unrealistically small, densely overlapping toy vocabulary (a 5-word cycle on a 5-letter alphabet), but the $n = 3$ figures are a genuine, measured answer to

”how significant a departure is it to skip the veto step?”: even a tiny 5–7-word vocabulary already produces an ambiguous grid 1 time in 12–25. This is direct evidence (not argument) that M_2 (with the veto step) is not an optional refinement – it is needed for M_1 to be a reasonable model of the real game at all, exactly as anticipated in the original design discussion.

11.3 Consequence: exact solving got much harder (but not intractable – correction 2026-09-18)

The corrected oracle is more informative per query (up to n^2 distinct outcomes – which cell – versus the old model’s effectively $\leq n + 1$), which sounds like it should make the game easier to solve. It does not make the *search* easier: richer outcomes mean a larger branching factor at every DP node, and the number of distinct reachable candidate sets explodes correspondingly. Concretely, at $n = 3$, excluded=‘a’ (5 candidate words, only 28,800 worlds – trivial under the old, wrong oracle, solved in < 1 s with ~ 100 memoised states): under the corrected oracle, **ExactSolver** passed 700,000 memoised states without terminating within 20 seconds. $n = 2$ remains fully tractable (re-solved below).

Correction, 2026-09-18: the claim above that this ”did not terminate” was wrong – it terminates in 26.5s at 767,832 memoised states, comfortably within reach; 20 seconds simply wasn’t tried for long enough before concluding intractability. This was an error of impatience, not of model or algorithm, but it is exactly the kind of claim this project exists to catch by testing rather than assuming – see Section 12.5 for the full, corrected $n = 3$ re-verification.

11.4 Consequence: the (word, line) sufficiency lemma is retracted

Section 5’s lemma held under the old oracle because a dross letter’s outcome (“absent”) genuinely didn’t depend on anything beyond (word, line). Under the corrected oracle, a dross letter’s revealed cell depends on the specific permutation of the remaining letters, which is *not* determined by (word, line) alone, and whose distribution over remaining cells stops being uniform across surviving pairs once some cells have already been revealed (precisely the complication that broke the very first F(L|G) attempt). **PairSolver** still silently implements the old model and is simply wrong now; its equivalence tests are kept in `tests/test_pair_solver.py` but marked `xfail/skip` with an explanation, as the documented record of the retraction rather than being deleted. A correct pair-based (i.e. not full-world) solver requires properly counting consistent completions per pair given the revealed history so far – not yet attempted.

11.5 Re-solved: $n = 2$, corrected oracle and termination

excluded	worlds	optimal $\mathbb{E}[\text{queries}]$	greedy $\mathbb{E}[\text{queries}]$
a	36	2.944	2.944
b	36	2.944	2.944
c	36	2.944	2.944
d	36	2.944	2.944
e	36	2.944	2.944

Greedy still matches optimal at $n = 2$ (as before the fix – still plausibly just an artefact of this toy instance’s size/symmetry; Section 7’s $n = 3$ counterexample was not re-verified under the corrected oracle because **ExactSolver** does not currently finish at $n = 3$ post-fix – see above). Note the value itself changed from the (wrong) 2.278 to 2.944: full-grid termination is a strictly

harder target than tagged-pair termination, so this increase is expected, not a sign of a new bug, *conditional on "know the whole grid" being the right definition of winning* – which is exactly the open question below.

11.6 Open question, not yet resolved: what does "winning" require?

Grid-uniqueness termination is *sound* (it can't loop forever, and it never claims certainty it doesn't have), but it may be stricter than necessary. The real game's actual win condition is presumably "the player can read off some line that is certainly a complete, valid word" – not "every dross cell is known." Those two conditions coincide throughout almost the entire game, and differ only in collision states, where "the designated word" is unknowable but "some certainly-correct line" might still not require full grid knowledge either way. A closer-to-real definition would be: stop as soon as some (w, p) is spelled by *every* remaining consistent grid (not necessarily the one tagged as "designated," and not necessarily requiring every other cell to be pinned down too). This was noticed, not yet implemented, and the 2.944 figure above should be read as an upper bound on expected queries under the real win condition, not necessarily the answer to the question we actually care about.

Resolved 2026-09-18 (see Section 12): neither the grid-uniqueness termination above nor the closer-to-real guess was right. There are three distinct quantities in play: (a) logical tag certainty (the candidate set collapses to one designated (w, p) pair by elimination) – but the real game does *not* stop here, it grants no early exit for merely knowing the answer; (b) logical grid certainty (all n^2 cells' values are deducible by elimination) – also not the real rule, and *also not simply "more than" (a) and (c) in the way first assumed here*; (c) the true line's own n cells *operationally, individually queried* (not merely deducible) – this is the real rule.

A same-day second correction, since the first attempt above (in the version of this paragraph originally written) also got it wrong: it claimed $(a) \leq (c) \leq (b)$ always. That is false, and Section 12 found the counterexample directly rather than by further armchair reasoning: at $n = 2$, grid-certainty (b) is measured at 2.09, strictly *below* mop-up (c) at 2.27. The reason is that (a) and (b) are both *deductive* quantities – reachable purely by elimination, without ever directly querying the cells they concern – while (c) is an *operational* one: mop-up specifically requires those n cells to have been individually, directly queried, with no shortcut via inference. Deduction can outrun the mandatory direct reveal or fall behind it depending on how much cross-referencing power the vocabulary structure gives elimination, so (b) and (c) are simply not ordered in general. What *is* always true, because it is a logical implication rather than an empirical race: $(a) \leq (b)$ (grid certainty implies tag certainty) and $(a) \leq (c)$ (completing mop-up implies you now know the tag too, but tag certainty could have arrived earlier by elimination). See Section 12's `TagIdentifySolver` for the solver that is actually valid to compare `MopUpSolver` against, and why `ExactSolver` (grid-based) is not.

12 2026-09-18: the real scoring rule, M_2 at $n = 2$, and $n = 3$ redone

Two points raised by review of the previous day's entry, both real and both implemented here, working entirely at $n = 2$ as requested (this section is deliberately thorough at that one size rather than rushing to scale up, per the request that prompted it).

12.1 The real objective: mop-up cost, not certainty

The real game does not end when the hypothesis space collapses to certainty. It ends only when the oracle observes that every cell of the (unique, under M_2) true line has been individually revealed by a direct query – so deducing the answer with 2 of its letters still untested still costs 2 more turns, spent confirming exactly those letters, before the game reports how many turns it took. Every previous objective in this logbook (Sections 6–9 and even the corrected Section 11) computed something else: expected queries to *certainty*, not to *game end*.

This matters strategically, not just numerically: a query landing on a true-line cell does double duty (it narrows the hypothesis space *and* advances the mandatory mop-up), while a query landing on dross only does the former. A policy optimising for fastest certainty need not be the policy optimising for fastest mop-up-inclusive game end.

Definition. $T :=$ the query count at which $\text{line}(w^*) \subseteq \text{Revealed}$, where w^* is whichever candidate is secretly true. Objective: minimise $\mathbb{E}[T]$.

12.2 M_2 : the veto, implemented and measured

M_2 is M_1 with the simplest veto policy already coded in `enumerate_worlds(..., reject_collisions=True)`: any grid whose dross letters happen to also spell a second dictionary word along some other line is discarded. Because M_2 never permits two valid words to coexist in the true grid, the earlier grid-uniqueness termination (Section 11) is no longer needed to avoid the identical-grid ambiguity: under M_2 , at most one candidate can ever have its own line fully covered by the true history at a time, so "some candidate's own line is now fully revealed" is unambiguous again.

Some (word, line) choices are unreachable under M_2 . A sharp observation from review: since M_2 discards any grid where the dross happens to spell a second word, it's possible for a specific (word, line) combination to have *zero* surviving completions – no arrangement of the leftover letters avoids a collision – and this can depend on which line, not just which word (a word banned from the bottom row might be fine on the top row, since the fixed cross-cutting cells differ). Measured directly for `WORDS_N2 = {ab, bc, cd, de, ea}` (all $5 \times 4 = 20$ possible (word, line, excluded) triples per word, 90 total across the vocabulary):

quantity	value
total (word, line, excluded) triples	90
triples with zero surviving M_2 completions	40 (44%)

44% is almost certainly inflated by this vocabulary's pathological structure (a 5-word cycle on a 5-letter alphabet, chosen originally for having "a little overlapping structure," which turns out to mean *a lot* at $n = 2$) – a less densely-overlapping toy vocabulary would presumably show a smaller rate, and the design discussion's own intuition was that this effect is "vanishingly unlikely" at the real game's $n = 5$. But the qualitative phenomenon is confirmed, not just plausible: entire (word, line) combinations really can be structurally impossible once uniqueness is enforced, and it really can depend on which line the word would occupy, not merely which word it is. (Verified by `test_some_word_line_choices_are_unreachable_under_m2`, and `test_m2_veto_actually_removes_collisions` confirms every surviving M_2 world genuinely spells exactly one dictionary word somewhere, not merely "at most one long the designated line".)

12.3 MopUpSolver: exact DP for the real objective

Implemented as `MopUpSolver` in `solver.py`. State is $(H, \text{Revealed})$ – candidate set *and* the revealed-cell set explicitly, not H alone: two different query histories can leave the same surviving-candidate set behind while having incidentally revealed different amounts of whichever line turns out to be true, and that difference changes T directly. (Cell identity alone suffices, without separately tracking which letter sits where: if a cell is already revealed, every surviving candidate is already forced to agree on its letter, by definition of "still consistent".)

$$V(H, \text{Rev}) = 0 \quad \text{if every candidate in } H \text{ has } \text{line}(w) \subseteq \text{Rev} \quad (3)$$

$$V(H, \text{Rev}) = \frac{|A|}{|H|} \left[1 + \min_L \sum_c \frac{|A_c|}{|A|} V(A_c, \text{Rev} \cup \{c\}) \right] \quad \text{otherwise} \quad (4)$$

where $A \subseteq H$ is the subset not yet "done" (own line $\not\subseteq \text{Rev}$) and A_c partitions A by the cell L resolves to. Already-done candidates contribute 0 to the expectation (their cost is already fixed by an earlier turn) via the $|A|/|H|$ scaling, and are dropped from further recursion. Unlike `ExactSolver`, a letter is never skipped merely for failing to discriminate between active candidates – it can still be valuable by advancing mop-up for all of them at once, and this objective must credit that.

One real bug caught while building this (not merely a design subtlety): allowing the solver to consider re-querying an already-revealed letter causes literal infinite recursion, not just wasted computation – the recursive call lands on the exact same (H, Rev) state before it is memoised. Already-revealed letters must be excluded from the letter search outright, not left for the argmin to (fail to) avoid.

Triple validation. `MopUpSolver`'s value at the root is checked against two independently-coded computations that share no formula with it or with each other: `simulate_mopup_policy_exact_expectation` (an independent recursive re-derivation) and `simulate_mopup_bruteforce` (literal step-by-step play against every one of the ≤ 36 worlds as ground truth, with no expectation formula at all – the closest thing available to "just play the game and see"). All three agree exactly, for every excluded-letter scenario.

12.4 Results at $n = 2$, M_2 , mop-up-aware objective

Correction, same day: "identify-only" below is `TagIdentifySolver` (tag/certainty-based termination), not `ExactSolver` (grid-based termination) as originally computed here – they are NOT interchangeable in general (see the correction to Section 11's resolved note, and Section 12.5 below for the counterexample that exposed this). At $n = 2$ specifically the two happen to coincide exactly (both 2.0909 for every excluded letter, confirmed directly), so the numbers below are unchanged – but that coincidence does not hold at $n = 3$, where they diverge substantially, so this table's agreement should not be read as "it doesn't matter which one you use."

excluded	M_2 worlds	tag-identify $\mathbb{E}[T]$	grid-identify $\mathbb{E}[T]$	mop-up-aware $\mathbb{E}[T]$
a	11	2.0909	2.0909	2.2727
b	11	2.0909	2.0909	2.2727
c	11	2.0909	2.0909	2.2727
d	11	2.0909	2.0909	2.2727
e	11	2.0909	2.0909	2.2727

The mop-up tax (over tag-identify, the only comparator with a proven ordering against mop-up) is exactly 0.1818 queries in every scenario for this vocabulary (identical across scenarios only because of this particular word list’s cyclic symmetry, not a general law). Note M_2 ’s hypothesis space is much smaller than M_1 ’s raw one – 11 surviving worlds here versus 36 under raw M_1 – a direct, visible consequence of the 44% impossibility rate above.

Optimal first move unchanged here, but don’t generalise from that. The mop-up-aware optimal first letter matches the identify-only optimal first letter in every scenario tested (e.g. excluded=‘a’: both pick ‘c’). That is a data point, not a proof that the two objectives’ optimal *policies* coincide throughout the game – $n = 2$ has only 2 letters per line and very little room for the double-duty effect described above to bite on move one specifically. Worth deliberately checking for divergence at $n = 3$, where a query has more structure to exploit, rather than assuming $n = 2$ ’s coincidence generalises.

12.5 $n = 3$, redone and confirmed complete

Prompted directly by ”n=3 next?”: rather than assume Section 7 and Section 11’s ”does not terminate” finding, it was re-tried with more patience, and the assumption was wrong.

$n = 3$ identify-only IS tractable from the opening – correction. ExactSolver on the full M_2 -filtered $n = 3$ opening state finishes in 26–42 seconds (760k–1.17M memoised states) depending on excluded letter, not “>20s, not observed to terminate” as Section 11 claimed. That claim was simply given too short a budget before being written down as a finding – a real methodological lapse (impatience presented as a result), corrected here by actually letting it run. Diagnostic: the branching factor really is $n^2 = 9$ distinct outcome cells per query at the root (measured directly), which is why 20s undersells it without being remotely infeasible.

excluded	M_2 worlds	optimal $\mathbb{E}[\text{queries}]$	greedy	solve time
a	27,420	6.0377	6.1411	26.5s
b	27,558	5.9990	6.1053	29.1s
c	26,922	6.0069	6.0685	28.9s
d	27,462	6.0126	6.0875	27.0s
e	27,144	5.9909	6.0680	26.0s
f	32,376	6.1005	6.2117	33.5s
g	32,126	6.0755	6.2046	33.9s
h	32,424	6.0907	6.1515	33.7s
i	32,100	6.0630	6.1561	33.6s
j	37,014	6.1770	6.2621	41.9s

Finding 2, strengthened. Greedy is strictly worse than optimal in *all ten* excluded-letter scenarios under the corrected oracle and M_2 (previously 6 of 10, under the buggy oracle and raw M_1) – not a weaker result than the original Finding 2, a stronger one. This is now the load-bearing confirmation that one-step maximum-entropy selection is not optimal play, on the actually-correct model, not the retracted one.

MopUpSolver at $n = 3$, against the correct comparator. The first attempt at this comparison used ExactSolver (grid-based) as ”identify-only” and got mopup < identify (4.02 vs 6.04) – apparently impossible, since mop-up can only add cost over mere certainty, never remove it. Triple-checking MopUpSolver again confirmed it was NOT the bug (all three independent checks still agreed exactly, even on a small fast sub-case). The bug was in the comparison itself: grid-based certainty is not a valid lower bound on mop-up cost at all (see the corrected note on Section 11’s

resolved paragraph, above) – only `TagIdentifySolver` (tag-based certainty) has a proven ordering against `MopUpSolver`. Redone with the right comparator:

excluded	M_2 worlds	tag-identify	mop-up-aware	tax
a	27,420	3.3700	4.0161	0.6461
b	27,558	3.3060	3.9792	0.6733
c	26,922	3.5938	4.2672	0.6734
d	27,462	3.5741	4.2423	0.6682
e	27,144	3.2555	3.8767	0.6212
f	32,376	3.5607	4.1069	0.5463
g	32,126	3.5576	4.1672	0.6096
h	32,424	3.4869	4.1172	0.6302
i	32,100	3.4886	4.1031	0.6145
j	37,014	3.8287	4.4244	0.5956

Every scenario correctly satisfies $\text{tag-identify} \leq \text{mop-up}$ (unlike the first, buggy-comparator attempt). The mop-up tax at $n = 3$ (0.55–0.67 queries) is roughly 3–4 $\times$ the $n = 2$ tax (0.18 queries) – weak evidence, from two data points, that the tax grows with n (more cells per line to mop up), but not something to extrapolate confidently from yet. `MopUpSolver` itself needed 45–80s per scenario here (3.46M– 4.53M memoised states), noticeably more than `TagIdentifySolver`’s 21–33s, consistent with the richer (H , Rev) state.

12.6 Status as of end of 2026-09-17/18 session

Items (i)-(iii) below were the open list at the end of the previous entry. All were subsequently addressed overnight 2026-09-18 (unattended, while the collaborator slept – see Section 13 for the write-up and for the resource-budget reasoning behind what was and wasn’t attempted).

- (i) check the whole optimal *policy* (not just the first move) for divergence between identify-only and mop-up-aware objectives – **done, Section 13: real, non-tie divergence confirmed at $n = 2$ and (sampled) at $n = 3$.**
- (ii) repeat the impossible-triple measurement with a less pathologically cyclic $n = 2$ vocabulary – **done, Section 13: confirmed vocabulary-density-driven, not an $n = 2$ -specific artifact.**
- (iii) attempt $n = 3$ under M_2 with the mop-up-aware objective – **done in Section 12.5 above**, ahead of this list’s original ordering, once tractability was confirmed by direct testing rather than assumed.

13 2026-09-18 (overnight): closing the two open flags

Done unattended overnight at the collaborator’s request, with an explicit instruction to pace the work against a visible resource budget (context window, 5-hour and weekly usage limits) rather than risk exhausting it or forcing an ungraceful auto-compaction mid-session. Given that constraint, the choice was deliberately to *firm up* the two flagged open questions from earlier the same day rather than open a new, higher-risk front (a completion-counting pair-based solver, or a first attempt at $n = 4$) – both of those are exactly the kind of undertaking that has produced subtle, hard-to-catch bugs each time it’s been tried so far in this project, and unsupervised overnight work is a bad time to be introducing new fragile machinery with no one to catch a wrong turn quickly.

13.1 (ii) Collision rate is vocabulary-density-driven, not an $n = 2$ artifact

Measured the impossible-triple rate (Section 12) against three more $n = 2$ vocabularies, alongside the original WORDS_N2 (44%):

vocabulary	words	impossible triples
sparse ($\{ab, cd, be\}$, minimal overlap)	3	7.4%
original WORDS_N2 (5-word cycle)	5	44.0%
denser ($\{ab, cd, ae, bc\}$)	4	36.1%
degenerate (every possible 2-letter string is "a word")	20	100.0%

The 100% figure for the degenerate case is not a bug, and is worth spelling out precisely because it looks alarming at first glance: if *every* string is a valid word, every line of every grid trivially "spells a word," so no grid can ever have exactly one – M_2 becomes unsatisfiable by construction, correctly reported as 100% impossible rather than crashing or silently returning an empty result. This is a useful sanity boundary: the measured rate spans the full range from "vocabulary sparse enough that M_2 rejects almost nothing" (7.4%) to "vocabulary dense enough that M_2 rejects everything" (100%), with the original 44% sitting well inside that range rather than being an outlier or a sign of a bug. **Conclusion: the effect is real and vocabulary-density-driven, confirmed rather than merely plausible, and not specific to $n = 2$'s geometry.**

13.2 (i) The mop-up-aware and identify-only policies genuinely diverge, not just occasionally tie

Method: solve MopUpSolver at the root (its own recursion tries every letter, so its memo covers every state reachable under full exploration, not just the states on its optimal path). For every memoised state where MopUpSolver still had a genuine choice (i.e. was not yet terminal), look up what TagIdentifySolver would have chosen at the same candidate set (ignoring states where TagIdentifySolver was *already* certain – comparing "what would you do" against a solver with nothing left to decide is not a meaningful comparison, it just reflects that mop-up still has mandatory queries left after certainty, which is the entire point of Section 12). Where the two solvers picked different letters, computed the exact mop-up cost of forcing TagIdentifySolver's choice instead, to separate genuine strategic disagreement from harmless tie-breaking (multiple equally-good letters, arbitrarily broken by iteration order).

$n = 2$ (**exhaustive, all 5 excluded-letter scenarios, all reachable states**): 135 states where both solvers faced a real choice; 65 disagreed (48%); **all 65 were genuine strategic divergences, zero were ties** – the worst single-state gap was a full extra expected query if the tag-optimal choice were followed instead of the mop-up-optimal one. This closes the "don't generalise from move 1 coincidence" caveat from Section 12 decisively: the two policies are not the same policy, they only happened to agree at the very first move for that particular (small, symmetric) vocabulary.

$n = 3$ (**excluded='a', sampled 3000 of ~3.4M memoised states**): kept deliberately bounded rather than exhaustive, both for overnight resource-safety and because exhaustively re-solving TagIdentifySolver from scratch at every one of millions of distinct candidate sets was not a responsible use of an unattended budget. Of the 3000 sampled states, 142 had both solvers still facing a genuine choice; **138 of those (97%) were real strategic divergences**, none were ties, worst single-state gap again exactly 1.0 extra expected query. Consistent with, and if anything stronger than, the $n = 2$ result – no sign that the divergence is an $n = 2$ artifact that washes out with more structure; if anything it looks more pervasive with more structure to exploit.

Conclusion for (i): the mop-up-aware and identify-only (tag-based) optimal policies are substantively different policies at every board size checked so far, not merely different at isolated states or only at the very first move. Any future heuristic or approximate solver for this game needs to be built and evaluated against the mop-up-aware objective specifically – optimising for fastest certainty and then treating the rest as ”obviously fine because you already know the answer” measurably leaves queries on the table.

13.3 What was deliberately NOT attempted overnight, and why

In order of what would otherwise have been next: a completion-counting pair-based solver (to escape full-world enumeration without the retracted Section 5 lemma), $n = 4$ under M_2 , and branch-and-bound pruning for `MopUpSolver`. All three require new design decisions with real room to get subtly wrong (as every previous attempt at something structurally new in this project has, at least once, before being caught) – exactly the kind of work this project’s own stated practice says should be checked against brute force *as it is built*, which is much harder to do well without a collaborator available to sanity-check an intermediate result. Left for a session where that’s possible again, rather than risking a plausible- looking but wrong result sitting unreviewed until morning.